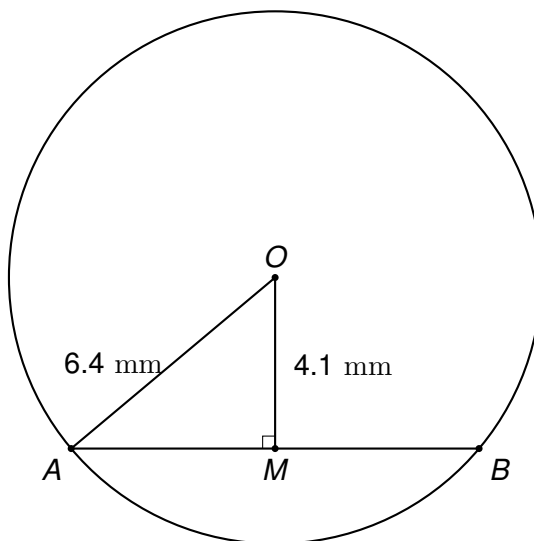


- 1 A biologist is examining a circular petri dish sample under a microscope. The circular boundary of the sample has centre O and radius 6.4 mm.

A straight bacterial colony forms a chord AB across the sample. The perpendicular distance from the centre O to the chord AB is 4.1 mm, meeting the chord at point M .



O is the centre of the circular sample, OA is a radius, and OM is perpendicular to the chord AB .

Calculate the length of the bacterial colony AB , giving your answer to 3 significant figures.

..... mm

(3)

(Total for Question 1 is 3 marks)
